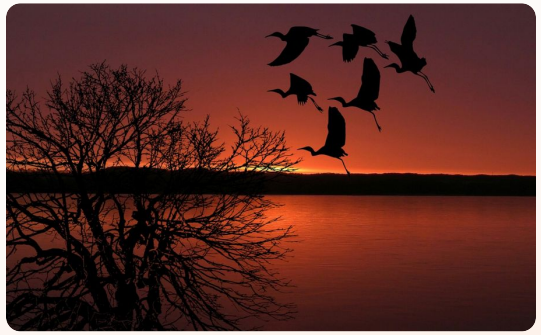
The Beauty of Atmospheric Layers

The Earth's atmosphere is divided into distinct layers that play a vital role in sustaining life, protecting us from harmful radiation, and creating awe-inspiring natural phenomena. Let's explore each layer and its unique characteristics.

By Om Jangam

The Troposphere



Lower Atmosphere

The layer closest to the surface of the Earth, where weather occurs, and where we live and breathe.

Atmospheric Boundary

Extends up to approximately 10km over the equator and is thinner at the poles. It contains 80% of the atmosphere's mass and most of its water vapor.

Thunderstorms

These intense storms can reach the upper Troposphere, unleashing lightning, hail, and strong winds.

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The Stratosphere

Ozone Layer

Protects us from harmful UV radiation by filtering out most of the sun's high-frequency waves.

Smooth Sailing

The Stratosphere is relatively calm with little turbulence or weather activity, making it ideal for air travel.

Above the Troposphere

Extends from about 10 to 50 km, with temperatures rising as you move higher up due to the presence of ozone.

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The Mesosphere

1

Meteoric Showers

Small rocks from space burn up in this layer to create the beautiful phenomena of meteoric showers.

2

Cool Temps

With temperatures as low as -100°C, this layer is the coldest in our atmosphere, and air molecules are highly dispersed.

3

Aurora Borealis

This layer, at heights of 50 – 85 km, is responsible for the stunning Northern and Southern Lights that grace our skies with a colorful display of light.

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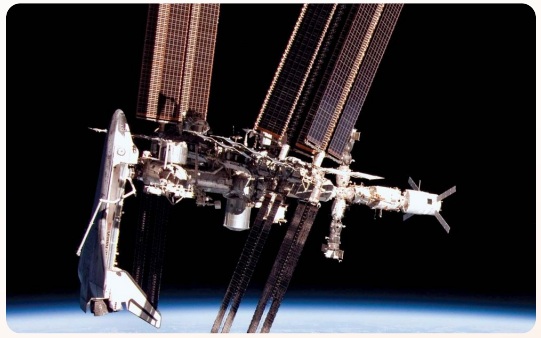
The Thermosphere



Temperature Increase

The upper atmosphere warms up here, due to the energy from the Sun's ultraviolet radiation.

Auroras

The upper part of the Thermosphere is responsible for auroras. Charged particles (ions) collide with gases, creating a spectacular light show.

Satellite Movement

Satellites orbit Earth at altitudes ranging from 160 to 2,000 km, located within this layer due to the low pressure we can see a few stars visible in the daytime from here.

By Om Jangam

The Exosphere

1 The Last Frontier

The Exosphere is the layer where space begins and atmospheric gases thin out until they are dispersed almost entirely.

2 Low Density

Neither atoms nor molecules are gas molecules there is no medium, so terrestrial debris left in this layer is not affected by air resistance.

3 Auroral zone

Upper Atmosphere study can use the auroral zone to probe both the lower ionosphere and the magnetosphere above it because the energy input into the atmosphere is easier to quantify via the auroras.

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Composition of Layers

	Troposphere	Stratosphere	Mesosphere	Thermosphere	Exosphere
Nitrogen	78%	78%	76%	73%	?
Oxygen	21%	21%	24%	25%	?
Argon	0.9%	0.9%	0.5%	?	?
Carbon Dioxide	0.04%	?	0.0001%	?	?

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Environmental Factors

Climate Change

The atmosphere is affected by global warming, leading to extreme weather events, rising sea levels and other climate-related issues. The greenhouse gases, namely, carbon dioxide, methane, and nitrous oxide, trap heat in the atmosphere, causing temperature increases.

Air Pressure

Each layer has its unique volume, temperature, and density, leading to different air pressures at different heights. Aeroplane cabins are pressurised to the equivalent of this layer.

Human Activities

Humans impact the atmosphere through pollution, deforestation, and other key activities. These actions can disrupt the delicate atmospheric balance and cause long-term damage.

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